

Topic 6: Image Generation Models

6.1 How Diffusion Models Work

- Diffusion models are the dominant approach for image generation (Stable Diffusion, DALL-E 3, Midjourney)
- Forward process (training): gradually add Gaussian noise to a real image over many steps until it becomes pure noise
- Reverse process (inference): train a neural network to predict and remove noise step by step, starting from random noise
- Text conditioning: a text encoder (often CLIP or T5) converts the prompt into an embedding that guides the denoising
- Result: starting from random noise, the model iteratively denoises towards an image that matches the text prompt

6.2 Key Image Generation Tools

- DALL-E 3 (OpenAI): integrated into ChatGPT Plus; excellent prompt adherence and ability to render text in images (openai.com)
- Midjourney: highest aesthetic quality for artistic and creative images; accessed via Discord or web (midjourney.com)
- Stable Diffusion: open-source model, free to download and run locally; huge ecosystem of community models (stability.ai)
- Adobe Firefly: trained on licensed Adobe Stock images -- safest for commercial use (firefly.adobe.com)
- Ideogram: strong at generating legible text within images; free tier available (ideogram.ai)
- Flux (Black Forest Labs): high-quality open-weight model, competitive with closed models

6.3 Writing Effective Image Prompts

- Describe what you want to see, not how to make it -- be specific about subject, style, lighting, mood
- Style keywords: 'photorealistic', 'oil painting', 'watercolour', 'isometric 3D', 'Studio Ghibli aesthetic'
- Lighting: 'golden hour', 'dramatic studio lighting', 'soft diffused window light', 'neon backlight'
- Camera/lens: '85mm portrait lens', 'wide-angle', 'aerial drone shot', 'macro photography'
- Negative prompts (Stable Diffusion): specify what to exclude -- 'no text, no watermark, no blur'

WEAK: a cat outside

STRONG:

A fluffy orange tabby cat resting on ancient stone steps,
purple wisteria in full bloom overhead,
soft golden-hour light, shallow depth of field,
85mm lens, photorealistic, award-winning photography

6.4 Stable Diffusion and Local Image Generation

- Stable Diffusion can be run entirely on your own hardware -- no usage fees, no content filters by default
 - Interfaces: Automatic1111 (feature-rich), ComfyUI (node-based workflow), Fooocus (simplified)
 - Hardware: NVIDIA GPU with 6GB+ VRAM recommended; runs on CPU but slowly
 - LoRA adapters: small fine-tuned files you add to the base model to change style or add a consistent character
 - ControlNet: control composition precisely using pose skeletons, depth maps, or edge maps as guides
 - Community models and guides: <https://civitai.com>
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